

STEPCONE-2025

STANDARD OPERATING PROCEDURE

EVENT NAME: CodeSpark Circuit Design

DEPARTMENT NAME: Electrical and Electronics Engineering (EEE)

EVENT TYPE: Technical Event

EVENT DESCRIPTION:

CodeSpark Circuit Design is a multi-round technical competition designed to push participants to apply both coding and electrical engineering principles. This event evaluates problem-solving skills, analytical thinking, and creativity through three key tasks: debugging error-filled code, analyzing incomplete circuits, and simulating designs using PSPICE. It fosters innovation, practical expertise, and real-world problem-solving while encouraging collaboration and technical mastery.

Round 1: Debugging Challenge

1. Participants are given five error-filled Python codes related to electrical engineering concepts.
2. Teams must identify and correct at least three codes within the allotted time.
3. The errors involve syntax, logical mistakes, or missing elements related to electrical calculations.
4. Teams are not allowed to modify the core logic of the code while debugging.
5. Teams that successfully correct the required number of codes qualify for Round 2.

Round 2:

1. Participants receive an incomplete circuit diagram along with the output specifications.
2. Teams need to calculate the missing component values (e.g., resistors, capacitors) based on the provided data.
3. The analysis must adhere to fundamental electrical principles and be completed within the time limit.
4. Teams must document their calculations and provide a final, corrected circuit diagram.
5. The top-performing teams advance to the final round based on the accuracy of

their analysis.

Round 3: Circuit Simulation Challenge

1. Teams must simulate their circuit design using industry-standard software like PSPICE.
2. The simulation should demonstrate that the circuit functions as intended and produces the expected behavior graph.
3. Participants must ensure their simulation is free of errors and accurately reflects their design calculations.
4. The simulation output, such as voltage and current graphs, must be submitted for evaluation.

EVENT GUIDELINES:

- Mobile phones are not allowed during the event.
- Participants must not use the internet during any round.
- Changes to answers/tasks are not permitted once submitted.
- Teams must manage their time effectively to complete tasks within the specified limits.

Team Size/No of Participants: 4 Members

PLAN OF ACTION: Day Shift

Duration of event:

3 hours – Day 2 - 8:00 AM to 12:00 PM

Judging Criteria:

Judging criteria include accuracy of debugged codes, correctness of circuit values, efficiency in task completion, clarity of circuit simulations, and overall problem-solving approach and methodology.